

Appendix Table A1. Full 12-Word IW Vocabulary

Word	Prevalence	Semantic label	Centroid description	Interpretation
IW00	3.02%	task-oriented right-hand approach	moderate body motion; brief/intermittent a balanced hands; right hand moves toward directed gaze; partner mostly passive	Task-oriented right-hand approach or placement while gaze remains task-directed.
IW01	59.79%	monitoring/listening	moderate body motion; brief/intermittent a balanced hands; partner-directed gaze; partner mostly passive	Dominant attention state; an attentive monitoring-like nonverbal state, not empty inaction
IW02	4.06%	left-hand withdrawal/action	high body motion; brief/intermittent active; hands; left hand withdraws; task-directed gaze; mostly passive	Task-focused left-hand withdrawal or completion after reaching, placing, or adjustment.
IW03	0.86%	rare partner-monitored right approach	high body motion; brief/intermittent active; dominant; right hand moves toward partner; partner-directed gaze; partner mostly passive	Rare salient right-hand approach with partner possible communicative reach or offer.
IW04	3.12%	strong task-focused right action	high body motion; brief/intermittent active; hands; right hand moves toward partner; task-directed gaze; partner mostly passive	Stronger task-focused right-hand action, likely manipulation or reaching in the shared workspace
IW05	1.61%	left-hand active with partner monitoring	high body motion; brief/intermittent active; dominant; partner-directed gaze; partner mostly passive	Left-hand-dominant active state while visually monitoring the partner.
IW06	6.67%	task-focused left withdrawal/action	moderate body motion; brief/intermittent a balanced hands; left hand withdraws; task-partner mostly passive	Common task-focused left-hand withdrawal/action often appears as an active response candidate
IW07	5.38%	task-focused right approach	moderate body motion; brief/intermittent a balanced hands; right hand moves toward directed gaze; partner mostly passive	Common task-focused right-hand approach with motion.
IW08	4.72%	active left-hand approach	high body motion; brief/intermittent active; dominant; left hand moves toward partner; partner-directed gaze; partner mostly passive	Important active response word: left-hand approach with partner-directed gaze.
IW09	1.01%	rare mixed transition	high body motion; brief/intermittent active; dominant; left hand withdraws; partner-directed gaze; partner mostly passive	Rare mixed action: right-hand dominant motion with withdrawal and partner gaze.
IW10	3.34%	partner-monitored right approach	moderate body motion; brief/intermittent a balanced hands; right hand moves toward partner; partner-directed gaze; partner mostly passive	Right-hand approach while gaze is partner-directed suggesting social checking or coordination.
IW11	6.40%	secondary monitoring/waiting	moderate body motion; brief/intermittent a balanced hands; partner-directed gaze; partner mostly passive	Secondary monitoring/waiting state that often precedes later active responses.